

Stage	What to Learn / Do	Free Resource / Book Link
1. Math Foundations	Linear Algebra, Probability, Statistics, Calculus	Mathematics for Machine Learning (Book, free PDF)
2. Python & ML Basics	Numpy, Pandas, Matplotlib, Scikit-learn	Python Data Science Handbook (free Jupyter version)
3. Classical ML	Regression, Trees, SVMs, Clustering, Evaluation	Andrew Ng's ML Course (Coursera, free audit)
4. Deep Learning Core	NN basics, CNNs, RNNs, Transformers, GANs	Deesp Learning Book (Goodfellow, free PDF) Dive into Deep Learning (interactive book)
5. Specializations	CV, NLP, RL, Generative AI	Stanford CS231n (Vision) Stanford CS224n (NLP) Spinning Up RL (OpenAI)
6. Practical Engineering (MLOps)	Deployment, Cloud, Pipelines	Full Stack Deep Learning (Course)
7. Research Skills	Reading & reproducing papers	ArXiv Sanity Preserver (paper explorer)
8. Projects & Competitions	Kaggle, GitHub, Blogs	Kaggle Learn (Free ML/DL tutorials)

Stage	What to Do (Job Focused)	Free Resource / Link
1. Strong ML/DL Skills	Finish IIT Ropar course + practice coding models in PyTorch/TensorFlow	Dive into Deep Learning
2. Build Portfolio Projects	Do 5–6 solid projects : - CV (image classifier, object detection) - NLP (chatbot, text classifier, LLM fine-tune) - Tabular ML (fraud detection, recommendation system) Deploy at least 2 on HuggingFace Spaces / Streamlit	Kaggle Datasets Streamlit HuggingFace Spaces
3. Competitions & Practice	Join Kaggle competitions to practice real-world ML problems	Kaggle Competitions
4. MLOps & Deployment	Learn: Docker, FastAPI, MLflow, Weights & Biases, basics of AWS/GCP	Full Stack Deep Learning Made with ML - MLOps
5. Resume + GitHub Proof	Keep GitHub active with projects + write short blog posts (Medium/Hashnode) explaining your work	Medium (Free blogging)
6. Networking	Contribute to open-source (HuggingFace, PyTorch) + LinkedIn presence	HuggingFace GitHub
7. Job Prep	Leetcode for DS/Algo basics + ML system design (how to scale ML in production)	LeetCode Free Problems System Design Primer